

Introduction to Knot Theory

(8) Slice Genus and 4-Dimensional Knot Topology

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Knots and Surfaces in B^4

Recall that the 3-sphere is the boundary of the 4-dimensional ball:

$$S^3 = \partial B^4.$$

A knot may bound surfaces inside B^4

$$\Sigma \subset B^4$$

with

$$\partial \Sigma = K.$$

A Seifert surface in S^3 can be pushed slightly into the interior of B^4 .

Slice Knot

Definition

A slice knot $K : S^1 \rightarrow S^3$ is a knot with smoothly embedded disk

$$D^2 \hookrightarrow B^4$$

such that

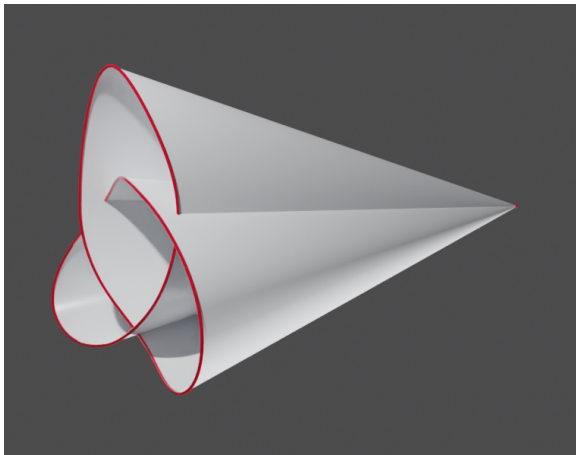
$$\partial D^2 = K.$$

In S^3 , the only knot bounding an embedded disk is unknot.

However, in B^4 , there are nontrivial knots bounding smoothly embedded disks.

Cone

There is a singularity, so cone cannot be slice.



https://en.wikipedia.org/wiki/Slice_knot

Slice Genus

Definition

The slice genus (or 4-ball genus) of a knot K is

$$g_4(K) = \min \{g(\Sigma) \mid \partial\Sigma = K, \Sigma \subset B^4\}.$$

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A knot K is slice if and only if $g_4(K) = 0$.

Recall

A Seifert surface of a knot $K \subset S^3$ is a compact orientable surface that satisfies

$$F \subset S^3 \quad \partial F = K.$$

The genus is similarly defined as

$$g_3(K) = \min \{g(\Sigma) \mid \partial\Sigma = K, \Sigma \subset S^3\}.$$

Unlike $g_3(K) = 0$, the condition $g_4(K) = 0$ does not determine a unique knot.

Comparison with Seifert Genus

Theorem

For every knot K ,

$$g_4(K) \leq g_3(K).$$

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For every knot K ,

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Push the interior of a Seifert surface $F \subset S^3$ slightly into the interior of B^4 .
Then F becomes a properly embedded surface in B^4 with the same genus.

Smooth vs Topological Slice

There are usually two types of slice.

- Smoothly slice: bounds a smoothly embedded disk in B^4
- Topologically slice: bounds a locally flat topological disk in B^4

In dimension 4, not all topologically slice are smoothly slice.

Unless we especially say for topologically slice, for this slice consider smoothly slice.

Theorem (Freedman, 1982)

A knot $K : S^1 \rightarrow S^3$ with Alexander polynomial of $\Delta_K(t) = 1$ is topologically slice.

Knot Concordance

Definition

Two knots $K_0, K_1 : S^1 \rightarrow S^3$ are concordant if there exists a smoothly embedded annulus

$$A \cong S^1 \times [0, 1] \subset S^3 \times [0, 1]$$

such that

$$\partial A = K_0 \times \{0\} \sqcup (-K_1) \times \{1\}.$$

Concordance can be viewed as a 4-dimensional deformation between knots.

Slice Knots and Concordance

Theorem

A knot K is slice if and only if it is concordant to the unknot.

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Suppose K bounds a smoothly embedded disk

$$D^2 \subset B^4.$$

Remove a small open 4-ball from the interior of B^4 .

The remaining manifold is diffeomorphic to

$$S^3 \times [0, 1],$$

and the disk becomes an annulus connecting K to the unknot.

The Knot Concordance Group

Concordance defines an equivalence relation on knots.

The set of concordance classes $\mathcal{C}$ forms an abelian group:

The group operation is connected sum:

$$[K_1] + [K_2] = [K_1 \# K_2].$$

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- Identity element: slice knots
- Inverse:

$-K$ = mirror image with reversed orientation

Ribbon Singularities

A ribbon singularity is a self-intersection of an immersed disk along an arc.

Locally, one sheet passes through another sheet, and the preimage of the singular arc consists of

- one arc in the interior of the disk,
- one properly embedded arc with endpoints on the boundary.

Ribbon Singularities

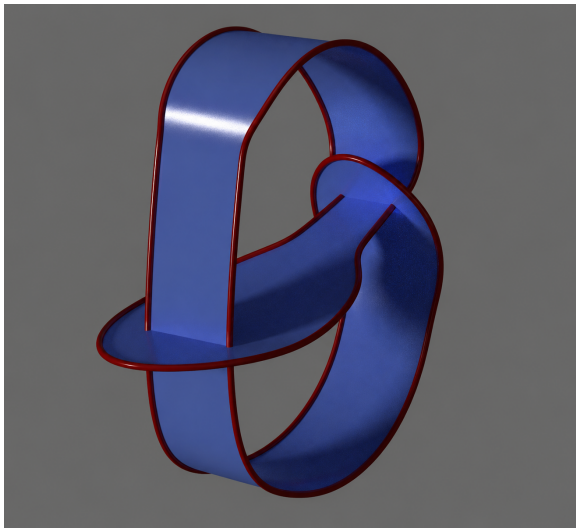


Figure: GPT-generated image

Ribbon Disks

A ribbon disk is an immersed disk

$$D^2 \looparrowright S^3$$

with all of its singularities are ribbon singularities.

Ribbon disks may have self-intersections, but only of the ribbon type.

Ribbon Knots

Definition

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A ribbon knot $K : S^1 \rightarrow S^3$ is a knot bounding a ribbon disk in S^3 .

Equivalently,

$$\partial D = K$$

for some immersed disk $D \subset S^3$ whose only singularities are ribbon singularities.

Ribbon Knots are Slice

A ribbon singularity can be removed by pushing one sheet slightly into the fourth dimension.

After removing every ribbon singularity,

$$D \subset B^4$$

becomes an embedded disk.

Hence

$$\partial D = K$$

and K bounds a smoothly embedded disk in B^4 .

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Theorem

Every ribbon knot is slice.

Recall : Problem C.1. of Week 3

Hint of Problem C.1.

Hint. 모든 매듭은 임의의 crossing에서 smoothing하면 component 수가 2인 link가 된다.

We can generalize this as a following.

After a single smoothing, the component of the link changes exactly $1(\pm 1)$.

We can say crossing change as a local change with two steps; smoothing and reverse of it.

Cobordism Inequality

Theorem

If two knots differ by one crossing change, then

$$|g_4(K_1) - g_4(K_2)| \leq 1.$$

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Idea.

A crossing change gives a genus one cobordism in

$$S^3 \times I.$$

Gluing this cobordism to a minimal genus surface for one knot produces a surface for the other knot. □

Cobordism Inequality

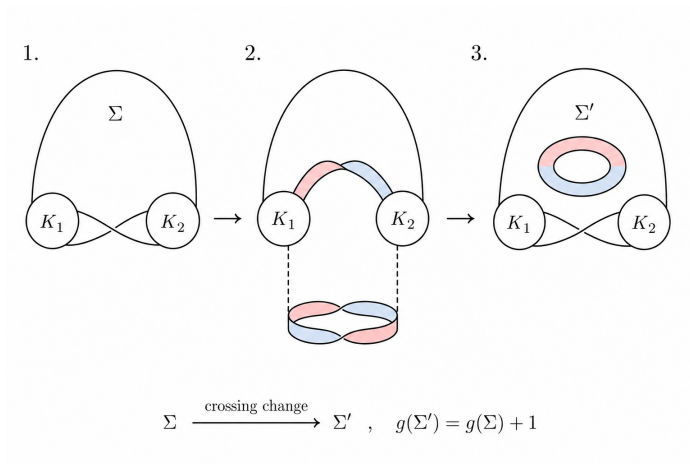


Figure: GPT-generated image

Kirby Calculus Interpretation

A crossing change can also be interpreted as ± 1 Dehn surgery on a small unknot surrounding the crossing.

In Kirby calculus:

- Framed circles correspond to 2-handles.
- Crossing changes arise from handle operations.

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Perspective	Interpretation
Surface cobordism	1-handle attachment
Kirby calculus	2-handle surgery

Knot Signature

Theorem

If two knots K_0 and K_1 are concordant, then

$$\sigma(K_0) = \sigma(K_1).$$

In fact, the signature induces a homomorphism

$$\sigma : \mathcal{C} \rightarrow \mathbb{Z},$$

whose image is the set of even integers $2\mathbb{Z}$.

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Corollary

If $\sigma(K) \neq 0$, then

$$[K] \in \mathcal{C}$$

has infinite order.

Proof Sketch

Recall:

$$[K_0] = [K_1] \iff K_0 \# (-K_1) \text{ is slice.}$$

Signature is additive under connected sum,

$$\sigma(K_0 \# (-K_1)) = \sigma(K_0) + \sigma(-K_1).$$

Also,

$$\sigma(-K_1) = -\sigma(K_1).$$

Therefore,

$$0 = \sigma(K_0 \# (-K_1)) = \sigma(K_0) - \sigma(K_1),$$

so

$$\sigma(K_0) = \sigma(K_1).$$

Image of the Signature Map

$$\sigma : \mathcal{C} \rightarrow \mathbb{Z}$$

is a group homomorphism because

$$\sigma(K \# J) = \sigma(K) + \sigma(J).$$

The image is exactly $2\mathbb{Z}$; remember that $\sigma(K) \in 2\mathbb{Z}$ obtained by $2g \times 2g$ matrix.

Murasugi's Inequality

Theorem (Murasugi, 1965)

For every knot K ,

$$|\sigma(K)| \leq 2g_4(K).$$

Equivalently,

$$g_4(K) \geq \frac{|\sigma(K)|}{2}.$$

Murasugi's Inequality

Theorem (Murasugi, 1965)

For every knot K ,

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Equivalently,

$$g_4(K) \geq \frac{|\sigma(K)|}{2}.$$

Hence a nonzero signature forces positive four-ball genus.

$$\sigma(K) \neq 0 \implies g_4(K) > 0.$$

Therefore K cannot be slice.

Recall : Khovanov Homology

Categorification

The Jones polynomial assigns a Laurent polynomial

$$V_K(q) \in \mathbb{Z}[q, q^{-1}]$$

to a knot K .

Khovanov homology upgrades this polynomial invariant to a homological invariant.

Jones Polynomial	$\implies$	Khovanov Homology
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Instead of assigning only numbers, we assign a bigraded chain complex whose homology contains richer topological information.

Recall : Khovanov Homology

Cube of Resolutions

Given a knot diagram with n crossings, each crossing is replaced by either its 0-resolution or 1-resolution (Kauffman smoothings).

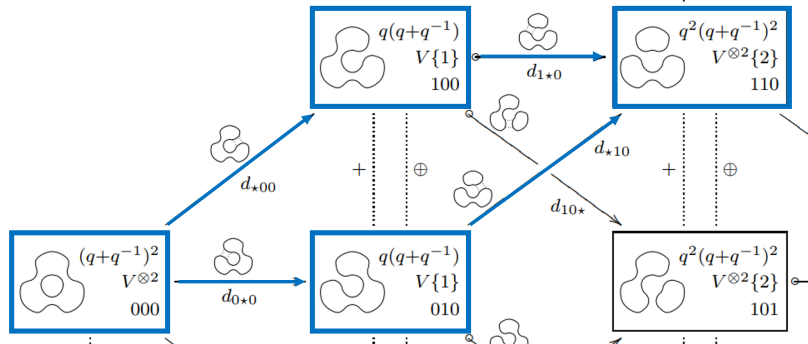
Each choice of smoothings

$$I = (i_1, \dots, i_n) \in \{0, 1\}^n$$

determines a collection of circles and hence a vertex of the n -dimensional cube.

- Vertices correspond to complete resolutions (collections of circles).
- Edges correspond to changing exactly one smoothing from 0 to 1.

Recall : Khovanov Homology



Frobenius Algebra

Definition

Let A be a finite-dimensional associative algebra over a field k .

A *Frobenius algebra* is an algebra equipped with a linear map

$$\varepsilon : A \rightarrow k$$

such that the bilinear form

$$\langle a, b \rangle = \varepsilon(ab)$$

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$$\varepsilon(ab) \neq 0.$$

The map ε is called the *counit*.

Recall : Khovanov Homology

Consider

$$A = \mathbb{Q}[X]/(X^2).$$

A basis is

$$\{1, X\}.$$

Define

$$\varepsilon(1) = 0, \quad \varepsilon(X) = 1.$$

Then

$$\langle a, b \rangle = \varepsilon(ab)$$

has matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which is nondegenerate.

Therefore A is a Frobenius algebra.

Multiplication and Comultiplication

The multiplication is

$$m : A \otimes A \rightarrow A$$

given by

$$1 \cdot 1 = 1, \quad 1 \cdot X = X, \quad X \cdot X = 0.$$

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The Frobenius structure determines a comultiplication

$$\Delta : A \rightarrow A \otimes A$$

by

$$\Delta(1) = 1 \otimes X + X \otimes 1,$$

$$\Delta(X) = X \otimes X.$$

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$$\Delta(X) = X \otimes X.$$

Thus A carries both algebra and coalgebra structures.

Recall : Khovanov Homology

Assigning Vector Spaces

For a resolution I containing $k(I)$ circles, assign the vector space

$$V(I) = V^{\otimes k(I)},$$

where

$$V = \mathbb{Q}[X]/(X^2).$$

Changing one smoothing corresponds to an edge of the cube of resolutions.

Recall : Khovanov Homology

Cochain Complex

For each homological grading

$$r = |I|,$$

define

$$C^r = \bigoplus_{|I|=r} V(I).$$

Recall : Khovanov Homology

Cochain Complex

For each homological grading

$$r = |I|,$$

define

$$C^r = \bigoplus_{|I|=r} V(I).$$

The edge maps induced by m and Δ assemble into a differential

$$d : C^{r,j} \rightarrow C^{r+1,j}$$

satisfying coboundary condition

$$d^2 = 0.$$

Thus we obtain a bigraded cochain complex

$$(C^{*,*}, d).$$

Recall : Khovanov Homology

Bigraded (Co)Homology

$$Kh^{r,j}(K) = H^r(C^{*,j})$$

- r : homological grading
- j : quantum grading

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Bigraded (Co)Homology

$$Kh^{r,j}(K) = H^r(C^{*,j})$$

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The graded Euler characteristic recovers the Jones polynomial:

$$\chi_q(Kh(K)) = J_K(q).$$

Lee Homology

Lee's Frobenius Algebra

Ordinary Khovanov homology uses

$$A = \mathbb{Q}[X]/(X^2).$$

Lee modifies the relation:

$$X^2 = 0 \quad \longrightarrow \quad X^2 = 1.$$

Thus the Frobenius algebra becomes

$$A_{\text{Lee}} = \mathbb{Q}[X]/(X^2 - 1).$$

This changes the multiplication and comultiplication maps, and therefore changes the differential.

Lee Homology

Modified Differential

The new differential has the form

$$d' = d + \Phi,$$

where

$$\deg(d) = 0, \quad \deg(\Phi) = 4.$$

Therefore, the quantum grading j changes.

$$\Phi : C^{r,j} \rightarrow C^{r+1,j+4}$$

Lee Homology

Filtered Complex

We can still induce a filtration:

$$\mathcal{F}^k C = \{x \in C \mid j(x) \geq k\}.$$

Thus Lee homology is naturally filtered homology.

Theorem (Lee, 2005)

For any knot K ,

$$\dim Kh_{\text{Lee}}(K) = 2.$$

After Lee deformation, all homology collapses to only two surviving generators.
For link, the connected components increases, so $\dim Kh_{\text{Lee}}(L) = 2^n$.

Lee Homology

Canonical Generators

For an oriented knot, Lee homology contains two canonical classes:

$$[s_+], \quad [s_-].$$

It is roughly defined from the filtration levels of the two canonical generators; these generators lie in nearby filtration levels.

The filtration difference between them encodes subtle geometric information.

Rasmussen used these filtration levels to define the invariant

$$s(K) \in \mathbb{Z}.$$

Rasmussen's s -Invariant

Let

$$[s_+], [s_-]$$

be the two canonical generators of Lee homology.

Although Lee homology is only filtered, each generator still carries a filtration grading.

Rasmussen defines

$$s(K) \in \mathbb{Z}$$

from these filtration levels by

$$s(K) = \frac{\text{qdeg}(s_+) + \text{qdeg}(s_-)}{2}.$$

Rasmussen's Slice Genus Bound

Theorem (Rasmussen, 2010)

For every knot K ,

$$|s(K)| \leq 2g_4(K).$$

Equivalently,

$$\boxed{\frac{|s(K)|}{2} \leq g_4(K)}$$

Therefore, the combinatorially defined invariant $s(K)$ gives a lower bound for the slice genus.

This was the first purely combinatorial method for studying smooth slice genus.

The Milnor Conjecture

For the torus knot

$$T(p, q),$$

the slice genus satisfies

$$g_4(T(p, q)) = \frac{(p-1)(q-1)}{2}.$$

Originally proved using gauge theory.

Rasmussen later proved it combinatorially using Khovanov homology.

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Theorems

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Images

- Wikipedia (Slice knot)